

Summer Research Project Report 2025

Salt Routes and Multicommodity Trade in the Western Himalayas

A Computational Historical Investigation

Submitted by

Samagra Bharti
2022115007
IIIT Hyderabad

Under the Supervision of

Dr. Aniket Alam
HSRC, IIIT Hyderabad

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1 Research Objective

This project investigates whether historical salt routes in the Western Himalayas functioned as the structural backbone of broader multicommodity trade networks. The research involves analyzing historical texts using computational methods to extract trade-related entities, themes, and relations, thereby modeling the interconnectedness of goods, places, and people across the region.

2 Methodology

2.1 Named Entity Recognition (NER) and Topic Modeling

Working on the previous NER model, I have integrated the Text Razor API to enhance the named entity recognition and topic identification from the corpus. The entities have been categorised along with their relevance and confidence scores, further capturing the high-level topics of each document. This two-dimensional scoring helps distinguish between core trade-related concepts (high relevance and confidence) and peripheral mentions (low relevance).

2.2 Confidence and Relevance Scores

Both confidence and relevance scores aim to evaluate the accuracy and importance of any extracted entity to a particular topic. The confidence score indicates how certain the model is that it has correctly identified the entity. This is especially useful in extracting rare or domain-specific entities such as *Chang-Pa* (people) or *Zanskar* (geolocation).

In contrast, relevance scores act as a tool to assess whether salt routes served as the backbone for multicommodity exchange, helping identify significant trade-related structures and practices. For example, isolating central themes such as specific commodities, trading groups, and tribes.

In this framework, high relevance and high confidence suggest strong evidence and are ideal for core analysis. High confidence but low relevance may indicate correct but minor observations and should be retained as metadata. High relevance with low confidence demands manual verification. The baseline values for these thresholds still need to be calibrated.

2.3 REBEL Relation Extraction

To move beyond named entity recognition and towards structured relation extraction, a custom pipeline was implemented using the REBEL model from Hugging Face. This pipeline extracts (Subject, Relation, Object) triples, focusing on commodity-related interactions, particularly those involving salt, wool, barley, and pashm.

However, the REBEL model is largely trained on general historical Wikipedia content and tends to be biased towards Euro-American topics. This makes the pipeline difficult to refine, and we may need to consider switching to another model or using a fine-tuned LLM.

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319 /biology/domesticated_animal
320
321 Camel (Relevance 0.261 | Confidence 6.479)
322 Cat (Relevance 0.351 | Confidence 8.964)
323 Columbidae (Relevance 0.312 | Confidence 4.927)
324 Dog (Relevance 0.339 | Confidence 11.050)
325 Donkey (Relevance 0.274 | Confidence 1.064)
326 Goat (Relevance 0.493 | Confidence 4.653)
327 Goose (Relevance 0.274 | Confidence 4.353)
328 Horse (Relevance 0.163 | Confidence 1.476)
329 Pony (Relevance 0.109 | Confidence 2.066)
330 Sheep (Relevance 0.443 | Confidence 6.397)
331 Yak (Relevance 0.454 | Confidence 10.020)
332
333 /biology/hybrid_parent_classification
334
335 Dog (Relevance 0.339 | Confidence 11.050)
336 Donkey (Relevance 0.274 | Confidence 1.064)
337 Horse (Relevance 0.163 | Confidence 1.476)
338 Lion (Relevance 0.155 | Confidence 7.888)
339 Pony (Relevance 0.109 | Confidence 2.066)
340 Tiger (Relevance 0.306 | Confidence 6.063)
341 Wolf (Relevance 0.584 | Confidence 10.930)
342
343 /biology/informal_biological_grouping
344
345 Eagle (Relevance 0.333 | Confidence 2.943)
346 Partridge (Relevance 0.117 | Confidence 4.744)
347
348 /biology/organism_classification
349
350 African wild ass (Relevance 0.319 | Confidence 2.145)
351 Alpine ibex (Relevance 0.090 | Confidence 3.580)
352 Animal (Relevance 0.268 | Confidence 1.342)
353 Antelope (Relevance 0.369 | Confidence 6.189)
354 Apple (Relevance 0.122 | Confidence 7.886)
355 Apricot (Relevance 0.285 | Confidence 6.347)
356 Argali (Relevance 0.601 | Confidence 1.628)
357 Bactrian camel (Relevance 0.242 | Confidence 13.770)
358 Bar-headed goose (Relevance 0.310 | Confidence 6.589)
359 Barley (Relevance 0.371 | Confidence 15.150)
360 Bear (Relevance 0.330 | Confidence 6.358)
361 Bearded vulture (Relevance 0.516 | Confidence 15.830)
362 Beaver (Relevance 0.306 | Confidence 6.806)
363 Bee-eater (Relevance 0.299 | Confidence 1.693)
364 Bharal (Relevance 0.425 | Confidence 13.090)
365 Birch (Relevance 0.139 | Confidence 7.423)

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Figure 1: Sample NER of topics with relevance and confidence scores.

2.4 On-going Work

Beyond NER, topic co-occurrence analysis has been initiated across a curated corpus comprising:

- *Becoming India* (Alam, 2019)
- *Ladakh: CrossRoads of High Asia* (Rizvi, 1996)
- *Simla Hill States* (1910)
- *Kangra District Gazetteer* (1897)
- *The Art of Not Being Governed* (Doe, n.d.)

From the NER results, high-frequency and high-confidence topics were selected, including salt routes, wool trade, and barley cultivation. For each salt-related term, a ± 50 -word context window was used to identify frequently co-occurring topics. This helped track the thematic proximity of ideas to salt, thus modeling their interrelation.

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1 /architecture/building
2 /architecture/structure
3 /biology/plant
4 /biology/animal
5 /boats/ship_type
6 /business/consumer_product
7 /business/business_location
8 /business/product_category
9 /business/product_ingredient
10 /business/product_line
11 /economy/livestock
12 /event/disaster_type
13 /food/food
14 /food/beverage
15 /food/ingredient
16 /fashion/fiber
17 /fashion/textile
18 /fashion/garment
19 /fashion/weave
20 /geography/mountain_range
21 /geography/mountain_pass
22 /geography/mountain
23 /geography/river
24 /geography/glacier
25 /geography/geographical_feature
26 /geology/geological_formation
27 /government/governmental_jurisdiction
28 /government/political_district
29 /location/location
30 /location/country
31 /location/in_state
32 /location/citytown
33 /location/in_district
34 /location/region
35 /location/administrative_division
36 /meteorology/forecast_zone
37 /meteorology/cyclone_affected_area
38 /metropolitan_transit/transit_line
39 /metropolitan_transit/transit_system
40 /organization/organization
41 /organization/organization_scope
42 /people/person
43 /people/profession
44 /people/ethnicity
45 /rail/railway
46 /religion/monastery
47 /religion/place_of_worship
48 /religion/religion
49 /religion/religious_leader
50 /religion/religious_practice

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Figure 2: Selected trade-related entity types extracted from the corpus.

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Anchor Entity: skardu [/location/location]
Example Context:
...stick; Chang-pa barter barley against salt with a Muslim villager from lower Ladakh 97 On the outskirts of Leh, a field i

Co-occurring Entities (sorted by frequency):
↳ ladakh [/travel/travel_destination] - 3 times
↳ polo [/sports/sport] - 2 times
↳ village [/fictional_universe/fictional_organization_type] - 2 times
↳ valley [/organization/organization] - 2 times
↳ gilgit [/location/statistical_region] - 2 times
↳ barley [/medicine/drug_ingredient] - 1 times
↳ suma [/people/person] - 1 times
↳ salt [/food/ingredient] - 1 times
↳ construction [/government/government_service] - 1 times
↳ house [/protected_sites/site_listing_category] - 1 times
↳ chang-pa [/people/person] - 1 times
↳ kashmir [/location/statistical_region] - 1 times
↳ trade [/organization/organization_sector] - 1 times
↳ srinagar [/travel/travel_destination] - 1 times
↳ landscape [/organization/organization_sector] - 1 times
↳ platoon [/military/military_unit_size] - 1 times
↳ pensi-la [/location/location] - 1 times
↳ jammu [/periodicals/newspaper_circulation_area] - 1 times
↳ time [/measurement_unit/dimension] - 1 times
↳ da-hanu [/people/person] - 1 times
↳ dras [/location/statistical_region] - 1 times

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Figure 3: Example of co-occurring entities around anchor entity Skardu.

3 Future Work

This pipeline needs further refinement, including expanding the commodity vocabulary, integrating spatial data, and better identifying traders and routes. Future goals include developing a commodity-route graph or visualizing salt route centrality. Extensive manual

reading of the corpus will also be needed to validate the automated extractions and refine model accuracy.